

Notes: Karabarbounis and Neiman (QJE, 2014)

The Global Decline of the Labor Share

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1 Big picture

Karabarbounis and Neiman document that the labor share of income, long treated as roughly stable in macroeconomic models, has declined globally since the early 1980s. Their preferred global corporate-sector measure falls by roughly 5 percentage points. The decline is broad: it appears in most countries, in most U.S. states, and in most major industries.

The paper argues that a large part of this decline can be explained by the falling relative price of investment goods. Investment goods became cheaper relative to consumption goods, plausibly because of investment-specific technological progress associated with information technology and the computer age. When capital becomes cheaper and the elasticity of substitution between capital and labor is above one, firms substitute toward capital strongly enough that labor's share of income falls.

The paper estimates the elasticity of substitution using cross-country and cross-industry variation in long-run trends in labor shares and relative investment prices. Its baseline estimate is about 1.25 to 1.30, comfortably above the Cobb-Douglas value of one. With this elasticity, the observed global fall in the relative price of investment explains roughly half of the global labor share decline. The rest may come from other forces, including rising markups, but the investment-price channel remains quantitatively central.

2 Incremental contribution of the paper

1. **Builds a new broad data set on corporate labor shares.** The paper compiles corporate-sector labor share data from country web pages, UN sources, OECD sources, and physical national-account books. Focusing on the corporate sector avoids the self-employment and unincorporated-business imputation problems emphasized by Gollin.
2. **Shows that the labor share decline is global and pervasive.** The decline is not just a U.S. or Western Europe fact. Most countries with sufficient data, most U.S. states, and most industries exhibit downward trends.
3. **Uses within-industry evidence to discipline explanations.** The within-industry component explains the vast majority of the labor share decline. This makes pure sectoral-composition stories, including simple trade reallocation stories, less plausible.
4. **Links the labor share decline to investment-specific technological change.** The paper connects the timing and cross-sectional incidence of labor share declines to the decline in the relative price of investment goods.
5. **Estimates substitution elasticity from cross-country trends rather than only one-country time series.** This identification strategy is important because the authors want to explain the common global decline. The estimate is identified from whether countries with larger investment-price declines also have larger labor-share declines.
6. **Separates the investment-price mechanism from markup, capital-augmenting technology, and skill-composition explanations.** The paper shows that allowing these forces does not eliminate the estimated elasticity above one or the contribution of cheaper capital.
7. **Highlights welfare implications.** A labor share decline caused by cheaper investment goods can be welfare-improving, while the same labor share decline caused by higher markups can be

welfare-reducing. The source of the decline matters.

3 Research question

The central question is: *why has the global labor share declined?* More specifically, the paper asks whether the decline can be explained by a standard production model once one allows the elasticity of substitution between capital and labor to exceed one.

The empirical target is the long-run trend in labor's share of value added. The mechanism is the long-run fall in the relative price of investment goods. If investment goods become cheaper, the rental cost of capital falls. With an elasticity of substitution $\sigma > 1$, capital and labor are sufficiently substitutable that firms increase capital relative to labor so strongly that the share of income paid to labor falls.

4 Data and measurement

4.1 Corporate labor share

The paper's preferred labor share measure is corporate compensation divided by corporate gross value added:

$$s_L = \frac{W^C N^C}{Q^C}.$$

This is attractive because the corporate sector does not require researchers to split proprietors' or self-employed income into labor and capital components. The corporate sector also avoids difficulties associated with specifying a production function for government activity.

The main data set includes 59 countries with at least 15 years of data between 1975 and 2012. The paper also uses U.S. state data and EU KLEMS industry data to check whether the trend is broad within regions and industries.

4.2 Relative price of investment

The relative price of investment is measured mainly from Penn World Tables and World Development Indicators. The authors also use KLEMS for industry-level analyses. The key fact is a sharp and widespread decline in the relative price of investment after around 1980.

The paper interprets this decline as investment-specific technical change: technological improvement in producing investment goods, especially information technology and computers, makes capital cheaper relative to consumption goods.

5 Model structure

The model has final consumption and investment goods produced from intermediate inputs. Investment goods become cheaper when the technology of producing investment goods improves

relative to consumption goods. Intermediate producers use capital and labor under monopolistic competition and charge a markup over marginal cost.

The key production function is CES:

$$Y_t = \left[\alpha_k (A_{K,t} K_t)^{\frac{\sigma-1}{\sigma}} + (1 - \alpha_k) (A_{N,t} N_t)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}},$$

where σ is the elasticity of substitution between capital and labor. The Cobb-Douglas case is the limiting case $\sigma = 1$.

A central implication is that the labor share depends on markups, capital-augmenting technology, and the rental rate of capital. Across steady states, if the household discount factor and the depreciation rate are constant, the change in the rental rate is tied to the change in the relative price of investment:

$$\hat{R} = \hat{\xi}.$$

Thus, long-run trends in the relative price of investment can be used as empirical variation in the rental cost of capital.

6 Baseline estimating equation

The benchmark empirical equation abstracts initially from changing markups and capital-augmenting technology. In that case, a linear approximation gives

$$\frac{s_{L,j}}{1 - s_{L,j}} \hat{s}_{L,j} = \gamma + (\sigma - 1) \hat{\xi}_j + u_j,$$

where j indexes countries or country-industries. A positive slope means $\sigma > 1$. If $\sigma = 1$, changes in the relative price of investment do not move the labor share. If $\sigma < 1$, cheaper capital would tend to raise rather than lower labor's share.

The authors identify σ from cross-sectional differences in long-run trends. This is a crucial design choice. The global downward trend in labor shares is not mechanically used to estimate the elasticity; instead, the paper asks whether countries or industries with larger investment-price declines also have larger labor-share declines.

7 Main findings

7.1 Labor shares declined broadly

The corporate labor share falls from roughly 64% to roughly 59% globally. The decline appears across the largest economies, across most countries, across most U.S. states, and across most industries. This is strong evidence against the old view of a stable labor share as a reliable macroeconomic constant.

7.2 Most of the decline is within industries

The KLEMS industry decomposition shows that more than 90% of the labor share decline comes from within-industry changes rather than shifts of activity across industries. This is important

because many trade or globalization stories predict between-industry reallocation. The paper does not deny trade matters, but the dominant fact requires a mechanism operating within industries.

7.3 The relative price of investment fell sharply

The relative price of investment goods declines dramatically after 1980, in all major data sources. This timing matches the acceleration in labor share declines. The decline in investment prices is broad enough to be a plausible global force rather than a country-specific shock.

7.4 Estimated elasticity exceeds one

The baseline estimates of σ are all above one, with an average of about 1.28. The preferred estimates imply that cheaper capital lowers the labor share because firms substitute toward capital strongly enough to reduce labor's income share.

7.5 Markup and skill alternatives do not overturn the result

Allowing for markups yields an average elasticity of about 1.26. Allowing different production nests for skilled and unskilled labor also gives an average estimate of about 1.26. Capital-augmenting technology could bias the estimate, but the paper argues the likely bias is small. Together, these checks show that the investment-price mechanism survives several alternative explanations.

7.6 The welfare interpretation depends on the cause

A decline in labor's share due to cheaper investment goods can increase output, consumption, and welfare. A decline caused by rising markups can reduce welfare. Thus the same observed labor share movement has opposite welfare implications depending on whether it reflects productive capital deepening or rising market power.

8 Extracted figures and tables from the paper

8.1 Figure I: Declining Global Labor Share

Read. Figure I plots the global corporate-sector labor share and the overall labor share after absorbing country fixed effects. Both series decline, with the corporate labor share falling from roughly 0.64 in the late 1970s to roughly 0.59 by the late 2000s and early 2010s.

Interpretation. This is the paper's motivating fact. The corporate series is especially important because it avoids the imputation of labor income for self-employed and unincorporated businesses. The figure shows that the decline is not a measurement artifact of one aggregate series: the overall economy series moves down as well. The pattern contradicts the common Kaldor-style assumption that factor shares are approximately constant in the long run.

8.2 Figure II: Declining Labor Share for the Largest Countries

Read. Figure II shows labor-share trends for the four largest economies: the United States, Japan, China, and Germany. All four have downward linear trends, although the timing and volatility differ. China has a visibly lower level of the labor share and a steep decline during the available period, while Germany has a pronounced downward trend from the 1980s onward.

Interpretation. The global decline is not driven only by small countries or by a single region. The largest economies, which account for a large share of global value added, move in the same broad direction. This figure is also useful because it shows heterogeneity: the mechanism must be broad enough to explain common decline, but flexible enough to allow different timing and levels.

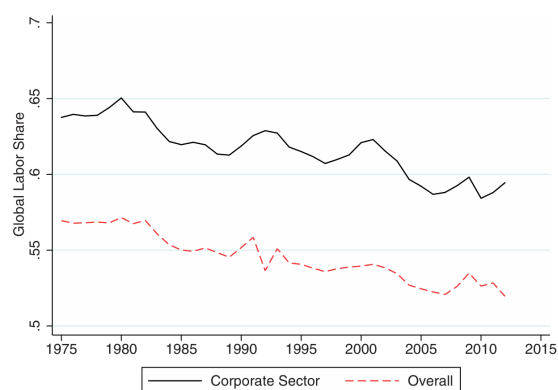


FIGURE I
Declining Global Labor Share

Figure I: Declining Global Labor Share

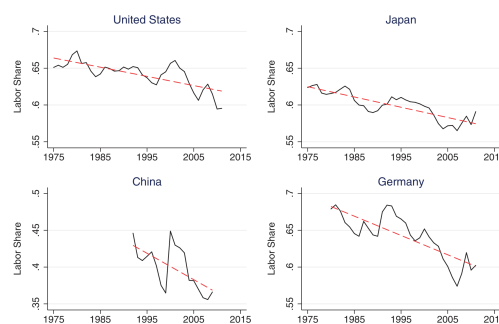


FIGURE II
Declining Labor Share for the Largest Countries

Figure II: Declining Labor Share for the Largest Countries

8.3 Figure III: Estimated Trends in Country Labor Shares

Read. Figure III reports country-specific linear trends in labor shares, scaled as percentage points per decade. Most countries lie below zero. The paper reports that 42 of 59 countries have downward trends, and among statistically significant trends, 37 are negative and only 9 are positive. The largest economies are highlighted.

Interpretation. This figure makes the word “global” credible. A weighted average could fall because a few large countries fall, but the country-by-country distribution shows pervasiveness. The large negative values on the left also show that some countries experienced very large declines, not just mild drift.

8.4 Figure IV: Estimated Trends in U.S. State Labor Shares

Read. Figure IV repeats the trend exercise within the United States using state-level data. Most states have negative labor-share trends, with the paper reporting declines in 34 states and increases in 17 states.

Interpretation. The within-U.S. evidence is an important robustness check. It shows that the decline is not solely a consequence of cross-country statistical practices, national institutions, exchange-rate weighting, or international sample composition. Even within one national accounting system and one labor market federation, the trend is broad.

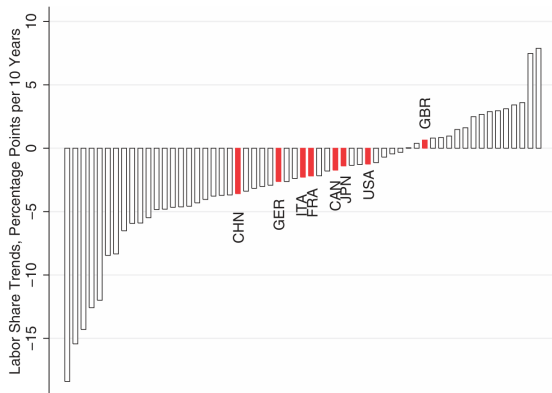


FIGURE III

Estimated Trends in Country Labor Shares

The figure shows estimated trends in the labor share for all countries in our

Figure III: Estimated Trends in Country Labor Shares

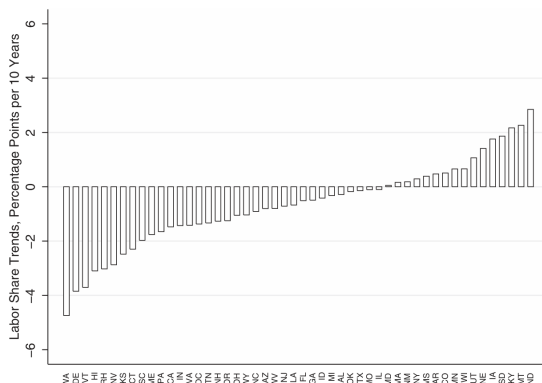


FIGURE IV

Estimated Trends in U.S. State Labor Shares

Figure IV: Estimated Trends in U.S. State Labor Shares

8.5 Figure V: Estimated Trends in Industry Labor Shares

Read. Figure V shows labor-share trends for 10 nonoverlapping industries in KLEMS. Mining, transport, manufacturing, and utilities have sizable negative trends. Agriculture and financial/business services show positive trends. The paper reports significant declines in six industries and significant increases in two.

Interpretation. The industry evidence has two messages. First, the decline is not limited to manufacturing. Second, there is heterogeneity across industries, which is useful for testing the theory because industries face different changes in investment prices and production structures. Manufacturing's decline is important, but it is only one part of a wider pattern.

8.6 Figure VI: Within Component and Total Trends in Country Labor Shares

Read. Figure VI plots each country's total labor-share trend against the within-industry component of that trend. Points cluster close to the 45-degree line. The paper concludes that more than 90% of the aggregate decline is attributable to within-industry changes.

Interpretation. This is one of the strongest pieces of evidence against a simple sectoral-composition explanation. If the labor share fell because economies moved from high-labor-share industries into low-labor-share industries, points would not line up so closely with the within-industry component. The result implies that the relevant mechanism works inside industries: firms or industries change factor usage, markups, or technology rather than merely reallocating activity across sectors.

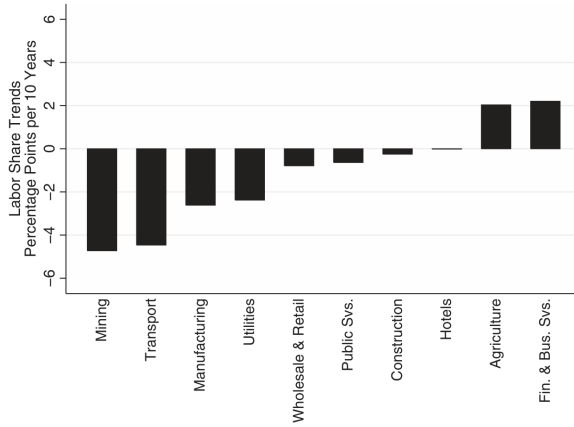


FIGURE V

Estimated Trends in Industry Labor Shares

Figure V: Estimated Trends in Industry Labor Shares

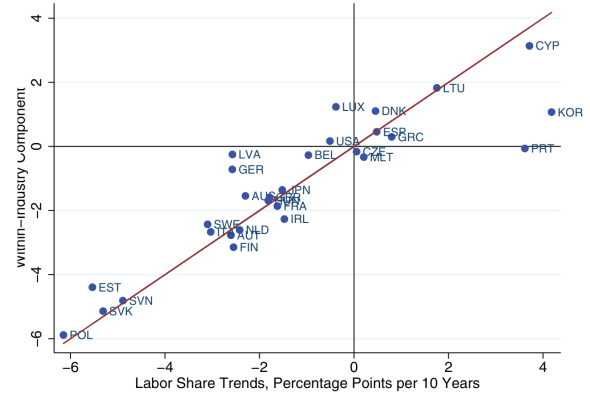


FIGURE VI

Within Component and Total Trends in Country Labor Shares

Figure VI: Within Component and Total Trends in Country Labor Shares

8.7 Figure VII: Declining Global Price of Investment Goods

Read. Figure VII plots the global log relative price of investment goods in PWT, WDI, and KLEMS sources. All series show a sharp decline after roughly 1980. The PWT series falls by around 0.3 log points from 1980 to the 2000s, while WDI and KLEMS show the same broad pattern.

Interpretation. This figure supplies the paper's candidate shock. The fall in investment prices means that capital becomes cheaper relative to consumption goods. Under $\sigma > 1$, firms substitute toward capital enough to reduce labor's share. The similar pattern across PWT, WDI, and KLEMS reduces concern that the investment-price decline is an artifact of a single data source.

8.8 Table I: Baseline Estimates of Elasticity of Substitution

Read. Table I estimates the elasticity of substitution under the baseline specification. Across country-level and KLEMS industry-level data sets, estimates range from 1.17 to 1.49, with an average of 1.28. The preferred country-level estimates are 1.25 using PWT investment prices and 1.29 using WDI investment prices.

Interpretation. Table I is the central quantitative result. Every estimate exceeds one, so the data favor a production structure where cheaper capital lowers the labor share. The result is robust across labor-share data sources and investment-price data sources. Since the equation is identified from cross-sectional trend variation, the table supports the claim that the investment-price mechanism explains part of the global decline rather than merely restating that both variables trend down globally.

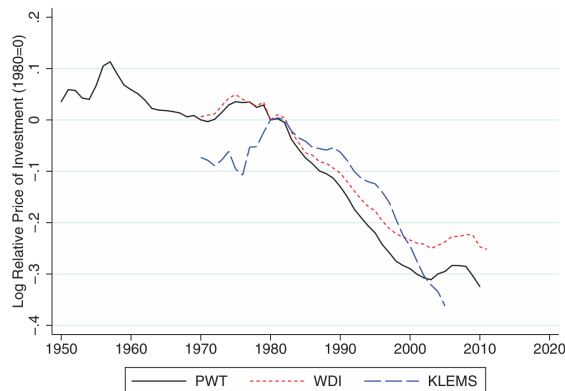


FIGURE VII
Declining Global Price of Investment Goods

Figure VII: Declining Global Price of Investment Goods

	Labor share	Investment price	$\hat{\sigma}$	Std. err.	90% Conf. interval	Obs.
(i)	KN Merged	PWT	1.25	0.08	[1.11,1.38]	58
(ii)	KN Merged	WDI	1.29	0.07	[1.18,1.41]	54
(iii)	OECD and UN	PWT	1.20	0.08	[1.06,1.34]	50
(iv)	OECD and UN	WDI	1.31	0.06	[1.20,1.42]	47
(v)	KLEMS 1	KLEMS	1.17	0.06	[1.06,1.27]	129
(vi)	KLEMS 2	KLEMS	1.49	0.13	[1.28,1.70]	129
Average			1.28			

Table I: Baseline Estimates of Elasticity of Substitution

8.9 Figure VIII: Labor Share and Relative Price of Investment

Read. Figure VIII plots the left-hand side of the estimating equation against trends in the relative price of investment. The fitted line slopes upward, with a slope around 0.28. Countries with larger declines in the relative price of investment tend to have larger normalized labor-share declines.

Interpretation. This scatterplot is the visual counterpart to Table I. It shows the cross-sectional identifying variation. A positive relationship means that countries where investment got cheaper faster experienced larger labor share declines, implying $\sigma > 1$. Importantly, the common global time trend is not what identifies the coefficient; the slope comes from country-level differences around that global trend.

8.10 Figure IX: Capital Share and Labor Share

Read. Figure IX plots trends in the log capital share against trends in the log labor share. The dashed 45-degree line shows the pattern that would occur if labor and capital shares moved together one-for-one. The fitted solid line is much flatter, with slope around 0.20 and a standard error of 0.23.

Interpretation. This figure addresses a markup-only story. If the labor share fell only because markups rose and factor cost shares were otherwise stable, labor and capital income shares would tend to fall together as profits rose. The near-flat relationship shows that this is not what happened. Markups may have increased, but they cannot be the entire explanation for the labor share decline.

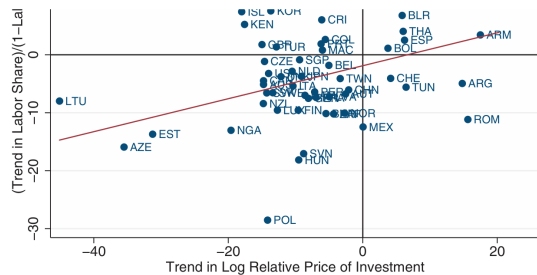


FIGURE VIII
Labor Share and Relative Price of Investment

The figure plots the left-hand and right-hand sides of equation (19). All values are scaled to denote changes per 10 years. For example, a value of -10 for the trend in the log relative price of investment means a roughly 10% decline of the price every 10 years. The figure excludes three countries (Kazakhstan, Kyrgyzstan, and Niger) with extremely low weights in the base-

Figure VIII: Labor Share and Relative Price of Investment

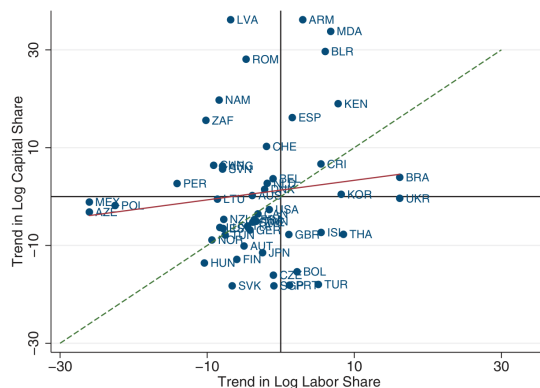


FIGURE IX
Capital Share and Labor Share

The figure plots the trend in the log capital share against the trend in the log labor share. All values are scaled to denote percent changes per 10 years. For example, a value of -10 for the trend in the log labor share means a 10% decline of the labor share every 10 years. For illustrative reasons, in this figure we drop three observations (Kazakhstan, Kyrgyzstan, and Macau) with extremely low weights in the regression of the first row of Table II and we winsorize one observation in each dimension for both variables. The solid line

Figure IX: Capital Share and Labor Share

8.11 Table II: Estimates of Elasticity of Substitution Allowing for Markups

Read. Table II modifies the elasticity estimation to allow markups to change. The estimates average 1.26, very close to the baseline average of 1.28. Some specifications are lower, such as 1.03 using KN merged labor shares, PWT prices, and corporate investment rates, but most remain clearly above one.

Interpretation. This table is important because rising markups are a plausible alternative explanation for a lower labor share. The results show that allowing for markups does not materially change the estimated elasticity or the contribution of cheaper investment goods. The paper's interpretation is not that markups are irrelevant; rather, markup changes matter but do not eliminate the investment-price channel.

8.12 Table III: Estimates of Elasticity of Substitution with Different Production Functions

Read. Table III allows different nests for skilled and unskilled labor relative to capital using KLEMS data. Estimates range from 1.19 to 1.34 and average 1.26. The estimates remain above one across specifications that nest high skill, middle and low skill, or low skill with capital.

Interpretation. This table addresses the concern that changing skill composition, skill premia, or capital-skill complementarity explains the results. Even when the production function distinguishes skill groups, the estimated substitution elasticity remains above one. Thus, the paper's mechanism is not simply a disguised skill-composition story.

TABLE II
ESTIMATES OF ELASTICITY OF SUBSTITUTION ALLOWING FOR MARKUPS

	Labor share	Investment price	Investment rate	$\hat{\sigma}$	Std. err.	90% Conf. interval	Obs.
(i)	KN Merged	PWT	Corporate	1.03	0.09	[0.87,1.19]	55
(ii)	KN Merged	WDI	Corporate	1.29	0.08	[1.16,1.42]	52
(iii)	OECD and UN	PWT	Corporate	1.24	0.11	[1.05,1.43]	46
(iv)	OECD and UN	WDI	Corporate	1.43	0.08	[1.28,1.57]	44
(v)	KN Merged	PWT	Total	1.11	0.11	[0.93,1.29]	54
(vi)	KN Merged	WDI	Total	1.35	0.08	[1.22,1.49]	52
(vii)	OECD and UN	PWT	Total	1.24	0.11	[1.06,1.43]	46
(viii)	OECD and UN	WDI	Total	1.42	0.09	[1.27,1.56]	44
Average				1.26			

Table II: Estimates of Elasticity of Substitution Allowing for Markups

TABLE III
ESTIMATES OF ELASTICITY OF SUBSTITUTION WITH DIFFERENT PRODUCTION FUNCTIONS

	Labor share	Nested input with capital	$\hat{\sigma}$	Std. err.	90% Conf. interval	Obs.
(i)	KLEMS 1	High skill	1.23	0.08	[1.11,1.36]	100
(ii)	KLEMS 1	Middle and low skill	1.19	0.08	[1.05,1.33]	100
(iii)	KLEMS 1	Low skill	1.19	0.09	[1.04,1.34]	100
(iv)	KLEMS 2	High skill	1.34	0.16	[1.07,1.60]	100
(v)	KLEMS 2	Middle and low skill	1.31	0.17	[1.03,1.60]	100
(vi)	KLEMS 2	Low skill	1.31	0.18	[1.02,1.61]	100
Average			1.26			

Table III: Estimates of Elasticity of Substitution with Different Production Functions

8.13 Table IV: Evaluating Labor Share's Decline (Percent Changes Across Steady States)

Read. Table IV compares steady-state responses in Cobb-Douglas and CES economies. A 25% fall in the relative price of investment has no labor-share effect in Cobb-Douglas, but it lowers the labor share by 2.6 percentage points in the CES economy. The same shock raises welfare equivalent consumption by 22.1% in the CES case versus 18.1% in Cobb-Douglas. A markup shock that lowers labor's share generates welfare losses. Combining investment-price and markup shocks in the CES case produces a 4.9 percentage point labor-share decline, close to the observed global decline, while still raising welfare by 15.8% because the cheaper-capital channel is beneficial.

Interpretation. Table IV is the paper's welfare message. A lower labor share is not inherently good or bad; its interpretation depends on the shock. If labor's share falls because investment goods become cheaper, households may benefit from higher capital accumulation, output, consumption, and welfare. If labor's share falls because markups rise, output and welfare fall. This distinction is central for policy: diagnosing the source of factor-share movements matters before drawing conclusions about welfare or regulation.

TABLE IV
EVALUATING LABOR SHARE'S DECLINE (PERCENT CHANGES ACROSS STEADY STATES)

	CD $\hat{\xi}$	CES $\hat{\xi}$	CD $\hat{\mu}$	CES $\hat{\mu}$	CD $(\hat{\xi}, \hat{\mu})$	CES $(\hat{\xi}, \hat{\mu})$
(i) Labor share (percentage points)	0.0	-2.6	-3.1	-2.6	-3.1	-4.9
(ii) Capital share (percentage points)	0.0	2.6	-1.9	-2.4	-1.9	-0.1
(iii) Profit share (percentage points)	0.0	0.0	5.0	5.0	5.0	5.0
(iv) Consumption	18.1	20.1	-5.2	-5.4	10.7	12.4
(v) Nominal investment	18.1	30.8	-11.1	-12.7	3.7	11.9
(vi) Labor input	0.0	-1.4	-3.2	-2.9	-3.2	-4.2
(vii) Capital input	51.6	67.8	-11.1	-12.7	33.2	43.6
(viii) Output	18.1	22.8	-6.3	-6.8	9.4	12.3
(ix) Wage	18.1	19.2	-8.2	-8.2	7.1	7.7
(x) Rental rate	-22.1	-22.1	0.0	0.0	-22.1	-22.1
(xi) Capital-to-output	28.4	36.6	-5.2	-6.4	21.8	27.9
(xii) Welfare equivalent consumption	18.1	22.1	-3.0	-3.4	13.2	15.8

Table IV: Evaluating Labor Share's Decline (Percent Changes Across Steady States)

9 How the argument fits together

The paper’s logic can be summarized in five steps.

1. The labor share has declined globally and within most countries, U.S. states, and industries.
2. The decline is mostly within industries, so explanations based only on sectoral composition are insufficient.
3. The relative price of investment goods fell sharply after 1980, making capital cheaper.
4. Cross-country and industry evidence implies an elasticity of substitution above one.
5. With $\sigma > 1$, cheaper investment goods induce capital-labor substitution large enough to reduce labor’s share; this mechanism explains about half of the observed decline.

10 Main assumptions and possible concerns

10.1 Steady-state interpretation

The paper treats long-run trends as movements from one steady state to another. This allows the authors to equate long-run rental-rate changes with relative investment-price changes. The assumption is more plausible for 15- to 37-year trends than for short-run fluctuations, but it abstracts from transition dynamics, financial frictions, and adjustment costs.

10.2 Constant discount factor and depreciation rate

The baseline assumes constant discount factors and depreciation rates over time. The authors check time-varying depreciation in KLEMS and find that the estimate changes little. Still, if unobserved capital-composition changes were large and correlated with investment-price trends, the rental-rate measurement could be affected.

10.3 Measurement of investment prices

Investment-price measurement is difficult because quality adjustment, especially for computers and IT capital, varies across countries. The paper uses multiple sources to reduce this concern. The similarity across PWT, WDI, and KLEMS is reassuring, but investment deflators remain imperfect.

10.4 Alternative explanations

The paper does not claim that the relative price of investment explains everything. Markups, technology bias, bargaining, institutions, globalization, and skill composition may also matter. The contribution is to show that the price of investment explains a large share of the decline and survives several robustness checks.

11 Key takeaways

1. The global labor share is not stable; it has declined substantially since the early 1980s.
2. The decline is pervasive across countries, U.S. states, and industries.
3. Most of the decline is within industries, not due to a shift toward low-labor-share industries.
4. The relative price of investment goods fell sharply over the same period.
5. The paper estimates an elasticity of substitution between capital and labor above one, around 1.25 to 1.30.
6. With $\sigma > 1$, cheaper investment goods reduce labor's share by inducing strong substitution toward capital.
7. Cheaper investment goods explain roughly half of the global labor share decline.
8. Rising markups and skill composition do not eliminate the investment-price mechanism.
9. Welfare implications depend on the cause of the labor share decline: cheaper capital is potentially welfare-improving, while higher markups are welfare-reducing.

12 How to use this paper

This paper is useful in macroeconomics, international macro, labor, and corporate income distribution debates. It provides a clean empirical fact, a tractable model-based explanation, and a reminder that factor-share movements are not mechanically welfare statements. For research design, the paper is also a good example of combining broad descriptive evidence, theory disciplined by an estimating equation, and robustness against several competing mechanisms.

The most important conceptual lesson is that the elasticity of substitution matters. If the production function is Cobb-Douglas, cheaper capital does not change factor shares. If capital and labor are more substitutable than Cobb-Douglas, investment-specific technological progress can permanently reduce labor's share.